

ECS Project Step 1 – Create ECS Cluster

To Begin with the Lab

This project step starts by creating an **Amazon ECS cluster** that supports both **Fargate** and **EC2**. The goal is to build a **hybrid cluster** so you can later compare launch types when creating tasks and understand how ECS behaves in real-world deployments. The PDF also shows the same implementation plan and cluster settings, including the default **Fargate** option and the EC2 configuration choices.

Summary of the Lab

In this step, an ECS cluster is created as the foundation for the full deployment project. The instructor explains that Fargate is included by default, but EC2 is also added so the project can test both launch types later. The cluster is configured with an Auto Scaling Group, ECS-optimized Amazon Linux 2023, T2 micro instances, an ECS instance role, and networking settings such as VPC, subnet, and security group. After the cluster is created, the EC2 instance becomes part of the ECS cluster infrastructure, and the Auto Scaling Group can be scaled later if needed. The instructor also shows a practical cost-control tip: if you want to stop using the EC2 instance temporarily, reduce the Auto Scaling Group desired capacity to zero so ECS terminates the instance automatically.

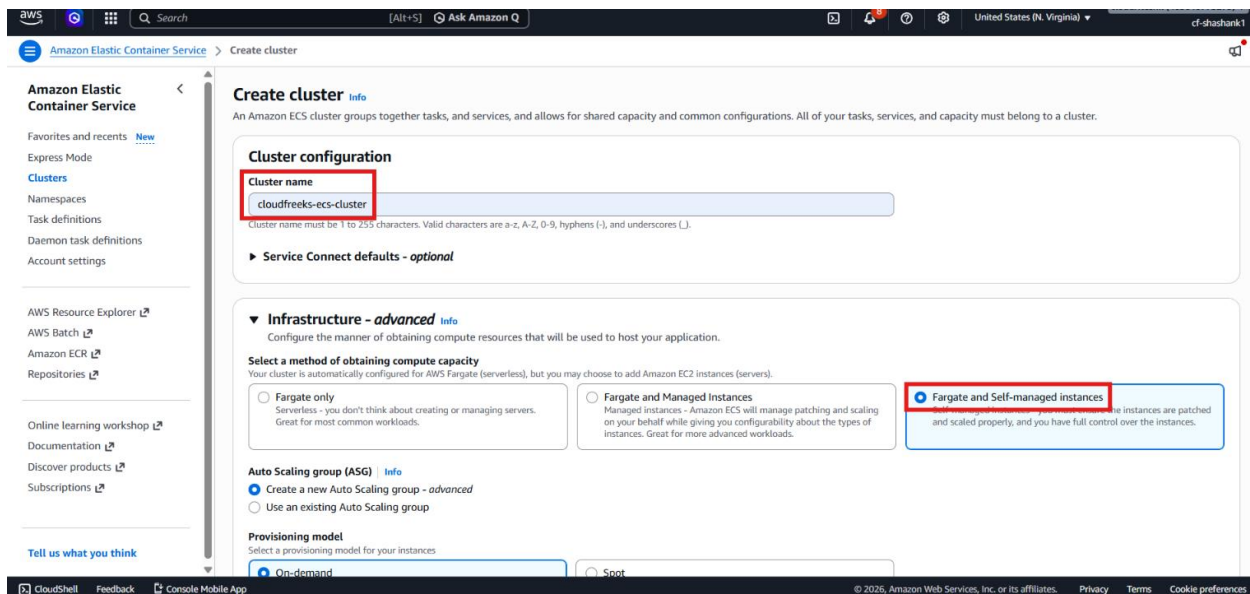
Creating the ECS Cluster

The purpose of this step is to prepare a **single ECS cluster** that can run workloads on both **Fargate** and **EC2**. This matters because later steps of the project will deploy the application in different ECS launch types, and a hybrid cluster makes that possible. The PDF on page 1 lists the project mission clearly: build a scalable PHP application using Amazon ECS and practice both launch types in the same environment.

Lab Steps

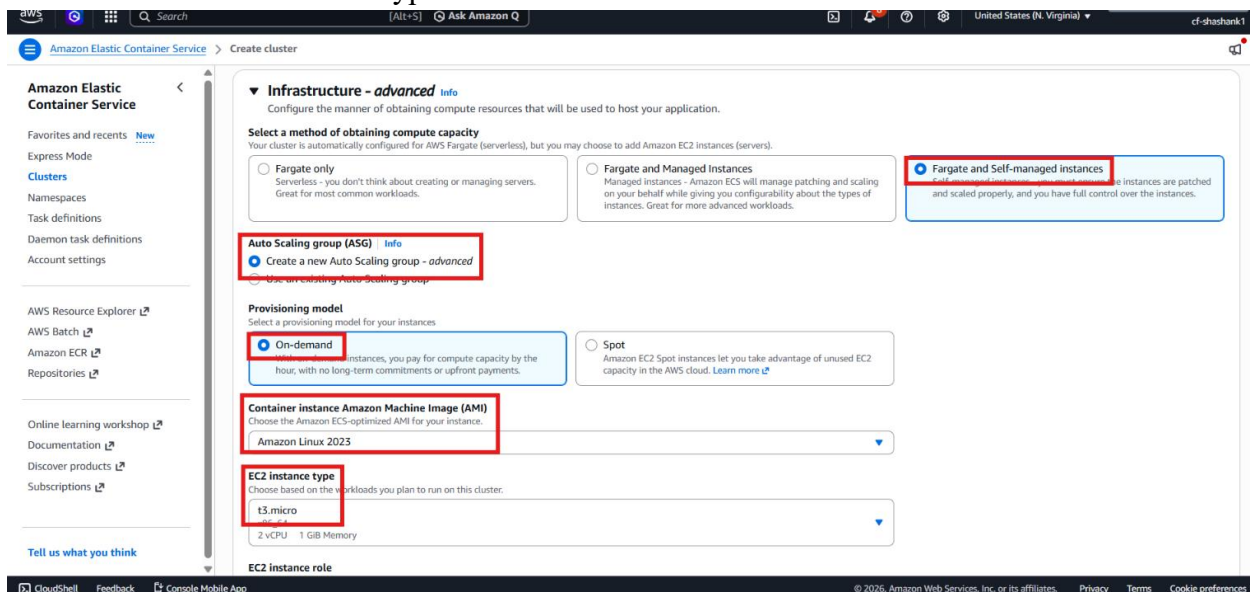
Step 1 – Open ECS and Start Cluster Creation

- Open the **Amazon ECS Console**.
- Go to **Clusters**.
- Click **Create Cluster**.
- Enter the cluster name: cloudfrecks-ecs-cluster

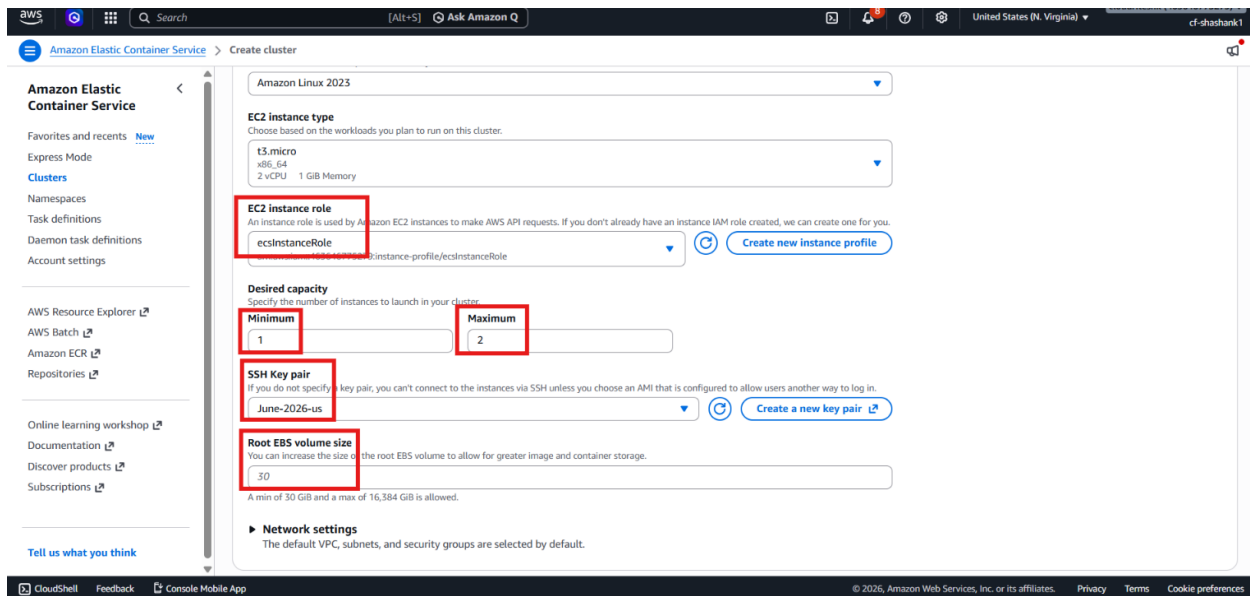


Step 2 – Configure the EC2 Infrastructure

- Select the option to add **Amazon EC2 instances as Self-Managed Instances**.
- Choose to create a **new Auto Scaling Group**.
- Set the provisioning model to **On-demand**.
- Select the ECS-optimized AMI: **Amazon ECS-Optimized Amazon Linux 2023**
- Choose the EC2 instance type: **t3.micro**

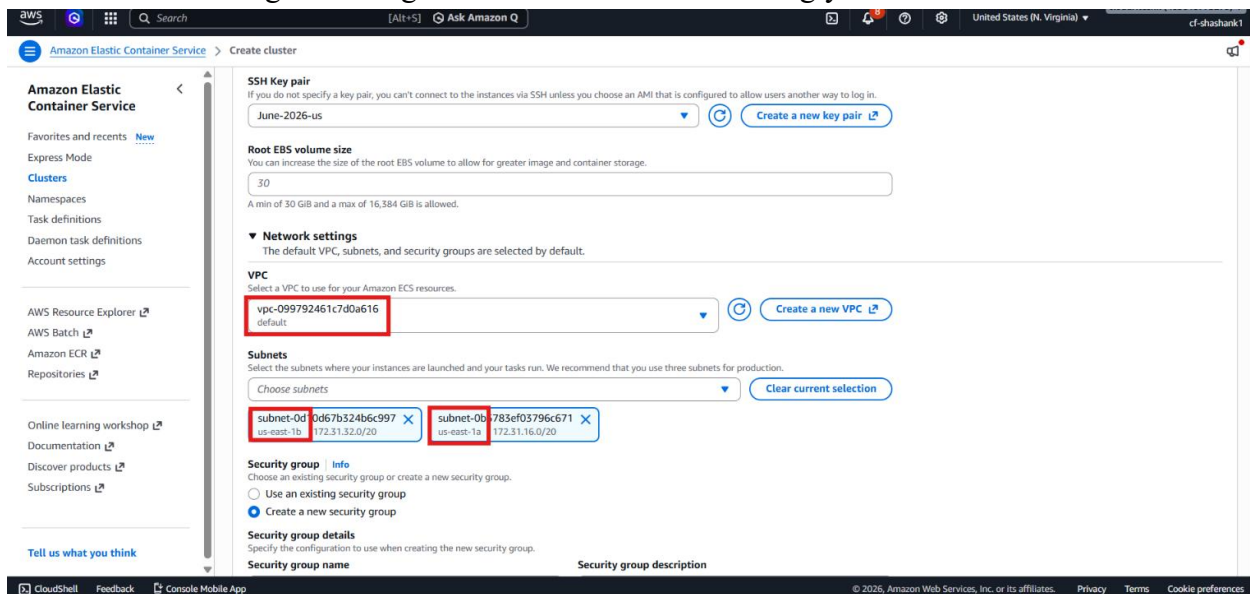


- Select or create the EC2 instance role.
- Set the desired capacity:
 - Minimum: 1
 - Maximum: 2
- Select your SSH key pair.



Step 3 – Configure Networking

- Use the **default VPC**.
- Keep the default subnet selection.
- If deploying in **Mumbai (ap-south-1)**, avoid using **ap-south-1c** for this free-tier setup because T2 micro instances may not always be available there.
- To avoid launch errors, deselect **ap-south-1c** if needed.
- But I am working in N. Virginia so I will subnets accordingly.



Step 4 – Create the Security Group

- Create a new security group.
- Use the security group name:

cloudfreeks-ecs-sg

- Add inbound rules:
 - **SSH** from 0.0.0.0/0
 - **Custom TCP 8080** from 0.0.0.0/0
- **Keep Auto-assign public IP set to Use subnet setting.**

Amazon Elastic Container Service > Create cluster

us-east-1b 172.31.32.0/20 us-east-1a 172.31.16.0/20

Security group Info

Choose an existing security group or create a new security group.

☐ Use an existing security group

☒ Create a new security group

Security group details

Specify the configuration to use when creating the new security group.

Security group name

cloudfreeks-ecs-sg

Security group name must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, underscores (_), hyphens (-), colons (:), forward slashes (/), parentheses (()), hashtrags (#), commas (,), at signs (@), brackets ([]), plus signs (+), equal signs (=), ampersands (&), semicolons (;), brackets ({}), exclamation points (!), dollar signs (\$), asterisks (*).

Security group description

Created in ECS Console

Security group description must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, underscores (_), hyphens (-), colons (:), forward slashes (/), parentheses (()), hashtrags (#), commas (,), at signs (@), brackets ([]), plus signs (+), equal signs (=), ampersands (&), semicolons (;), brackets ({}), exclamation points (!), dollar signs (\$), asterisks (*).

Inbound rules for security groups

Add one or more ingress rules for your security group.

Type	Protocol	Port range	Source	Values	
SSH	TCP	22	Custom	0.0.0.0/0	Delete
Custom TCP	TCP	8080	Custom	0.0.0.0/0	Delete

Enter a valid port or port range between 0 and 65535. For example: 80 or 0-1023.

[Add rule](#)

Auto-assign public IP Info

Choose whether to auto-assign a public IP to the Amazon EC2 instances

☐ Use subnet setting

[Tell us what you think](#)

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Step 5 – Create the Cluster

- Review all settings.
- Click **Create**.

Amazon Elastic Container Service > Create cluster

Monitoring - optional

Configure observability, encryption, and logging options to maintain compliance and operational visibility of your container environment.

Encryption - optional

Choose the KMS keys used by tasks running in this cluster to encrypt your storage.

Tags - optional Info

Tags help you to identify and organize your clusters.

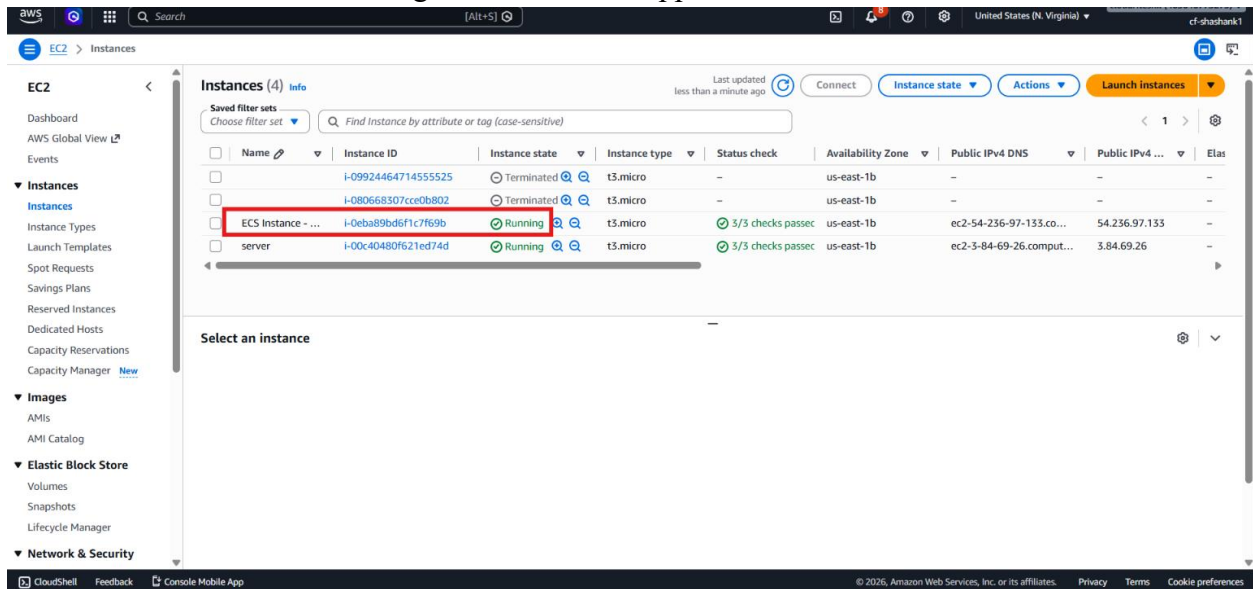
[Cancel](#) [Create](#)

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- Wait for ECS to create the cluster and the EC2 infrastructure.
- Confirm that the Auto Scaling Group launches the EC2 instance automatically.

Step 6 – Verify the EC2 Instance

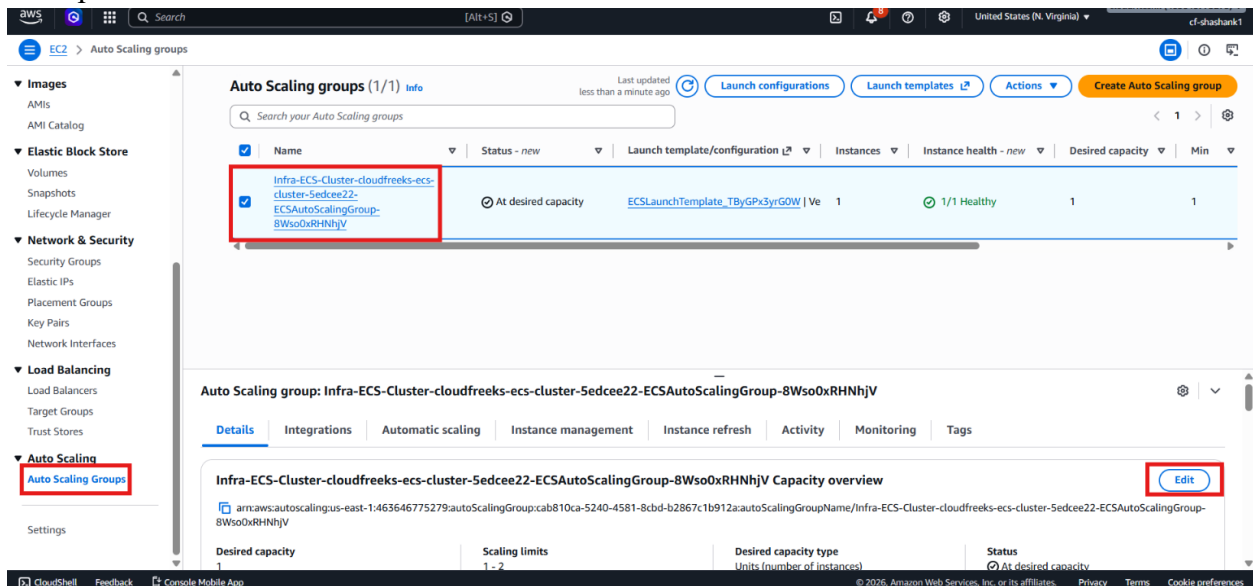
- Open the **EC2 Console**.
- Confirm that the ECS-managed EC2 instance appears.



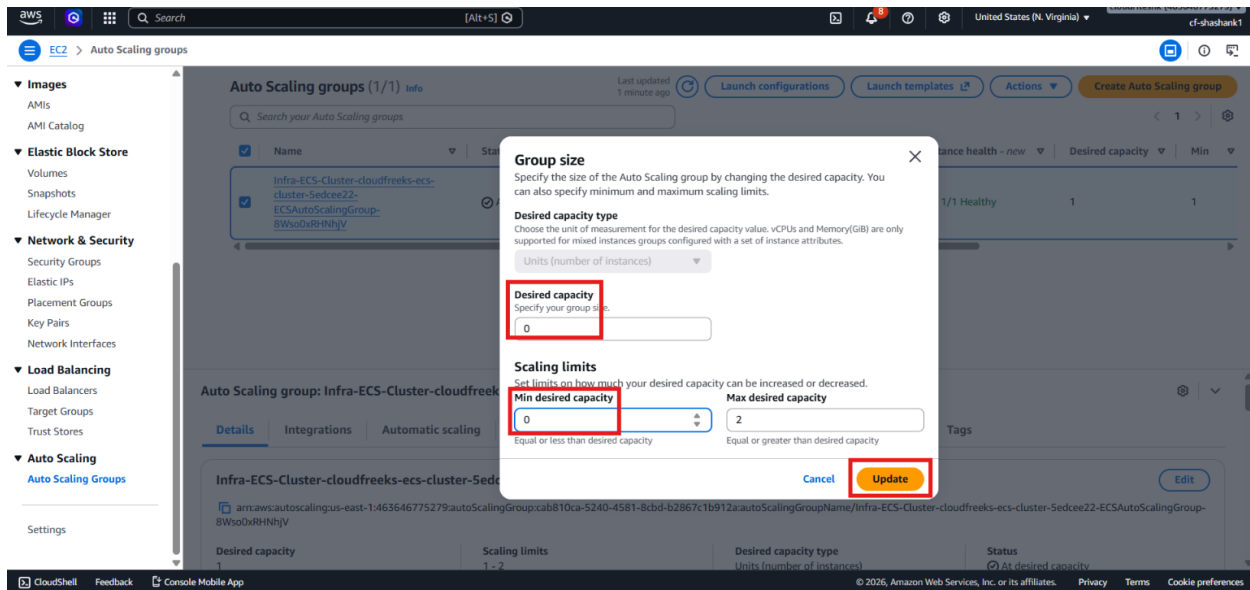
- Verify that the instance is associated with the ECS cluster.
- Confirm that the instance was created by the Auto Scaling Group, not manually.

Step 7 – Control Cost by Adjusting Auto Scaling

- If you want to pause the lab and avoid charges, go to the **Auto Scaling Group**.
- Open **Edit**.



- Change the desired capacity and min capacity to **0**.
- ECS will terminate the instance automatically.



- When you want to continue the lab, set the desired capacity back to **1**.
- ECS will create the instance again in a few minutes.

Verification

- Verify that the ECS cluster was created successfully.
- Verify that the cluster includes **Fargate** and **EC2** infrastructure.
- Verify that the EC2 instance is running as part of the ECS cluster.
- Verify that the Auto Scaling Group created the instance.
- Verify that the security group includes **SSH** and **8080** inbound rules.
- Verify that setting the Auto Scaling Group desired capacity to 0 terminates the instance.